PROBLEM 1.5

Forensic Time of Death Estimation (Algor Mortis)

Original statement · ProblemsCh1.pdf, p. 2

Algor Mortis • Thermal Inversion • Newton Cooling

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** A body is found in a cold room ($T_A = 5^\circ\mathrm{C}$) at 3 pm, and its temperature then is $19^\circ\mathrm{C}$. An hour later its temperature has dropped to $15^\circ\mathrm{C}$. Use Newton's law of cooling to estimate the time of death, assuming that the living body temperature is $37^\circ\mathrm{C}$ and measuring time in hours.

PHASE 2: FUNDAMENTAL FORMULATION

Let $t = 0$ h correspond to 3:00 pm. Known data: $T_A = 5^\circ\mathrm{C}$, $T(0) = 19^\circ\mathrm{C}$, and $T(1) = 15^\circ\mathrm{C}$ (at 4:00 pm). We seek the negative time $t_d < 0$ when $T(t_d) = 37^\circ\mathrm{C}$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: Solution with the integrating factor of the class notes [Notes, p. 3]. In standard form $T' + kT = 5k$: the linear equation $x' + p(t)x = q(t)$ with $p = k$, $q = 5k$, $t_0 = 0$, $T(0) = 19$.

1. **Find I.** $P(t) = \int_0^t k\,d\tau = kt$, so $I(t) = e^{kt}$ and $I(0) = e^0 = 1$.
2. **Integrate from 0 to t.** $\int_0^t \left(e^{k\tau}T(\tau)\right)' d\tau = \int_0^t 5k\,e^{k\tau}\,d\tau$. On the right substitute $s = k\tau$, $ds = k\,d\tau$: $5\int_0^{kt} e^s\,ds = 5\left[e^s\right]_0^{kt} = 5\left(e^{kt} - 1\right)$.
3. **Simplify.** Barrow's rule on the left: $e^{kt}T(t) - I(0)T(0) = e^{kt}T(t) - 19$, so $e^{kt}T(t) = 19 + 5e^{kt} - 5$. Dividing by e^{kt}: $T(t) = 19e^{-kt} + 5 - 5e^{-kt}$.

$$T(t) = 5 + 14e^{-kt} \tag{1.5.1}$$

This is the formula of the notes, $x(t) = \frac{x_0}{I(t)} + \frac{1}{I(t)}\int_{t_0}^t I(\tau)q(\tau)\,d\tau$. Check: $T' = -14ke^{-kt} = -k(T-5)$ and $T(0) = 19$.

Step 2: Determination of k:

At $t = 1$ h, $T(1) = 15^\circ\mathrm{C}$:

$$15 = 5 + 14e^{-k} \implies 10 = 14e^{-k} \implies e^{-k} = \frac{10}{14} = \frac{5}{7}$$

Inverting both sides:

$$e^{k} = \frac{7}{5} = 1.4 \implies \mathbf{k} = \ln(\mathbf{1.4}) \approx \mathbf{0.336472}\ \mathrm{h}^{-\mathbf{1}}$$

Step 3: Calculating Time of Death t_d:

Enforce the living core body temperature $T(t_d) = 37^{\circ}\mathrm{C}$:

$$37 = 5 + 14e^{-kt_d} \implies 32 = 14e^{-kt_d} \implies e^{-kt_d} = \frac{32}{14} = \frac{16}{7}$$

Take the natural logarithm:

$$-kt_d = \ln\left(\frac{16}{7}\right) \implies t_d = -\frac{\ln(16/7)}{k} = -\frac{\ln(16/7)}{\ln(1.4)}$$

Evaluating numerically with $\ln(16/7) \approx 0.8266786$ and $\ln(1.4) \approx 0.3364722$:

$$t_d = -\frac{0.8266786}{0.3364722}\ \mathrm{h} \approx -\mathbf{2.4569}\ \text{hours}$$

Step 4: Clock Time Conversion:

Multiply the fractional part by 60 minutes:

$$2.4569\ \text{hours} = 2\ \mathrm{h} + (0.4569 \times 60)\ \mathrm{min} = 2\ \mathrm{h}\ 27.41\ \mathrm{min} \approx 2\ \text{hours and}\ 27\ \text{minutes}$$

Subtracting 2 hours and 27 minutes from 3:00 pm (15:00):

$$15:00 - 2\ \mathrm{h}\ 27\ \mathrm{min} = \mathbf{12}:\mathbf{33}\ \mathrm{pm}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$t_d \approx -2.457 \text{ hours} \implies \text{Time of Death} \approx \mathbf{12} : \mathbf{33} \text{ pm}$$

Forensic Consistency: Algor mortis cooling is monotonic and deterministic under steady ambient conditions.